

ENGINE EMERGENCY CONDITIONS

3.3 ENGINE EMERGENCY CONDITIONS

3.3.1 Single Engine Failure - Hover IGE

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

- | | |
|----------------------------|---|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Landing attitude | – Establish |
| 3. Collective lever | – Raise as necessary to stop descent and cushion landing |

After landing:

- | | |
|---|------------------|
| 4. Single engine emergency shut-down | – Perform |
|---|------------------|

ENGINE EMERGENCY CONDITIONS

3.3.2 Single Engine Failure - Hover OGE

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

- | | |
|---------------------|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
|---------------------|---------------------------------|

- | | |
|-------------|------------------------|
| 2. Airspeed | – Increase if possible |
|-------------|------------------------|

● FORCED LANDING

- | | |
|---------------------|-------------|
| 3. Landing attitude | – Establish |
|---------------------|-------------|

- | | |
|---------------------|--|
| 4. Collective lever | – Raise as necessary to stop descent and cushion landing |
|---------------------|--|

After landing:

- | | |
|--------------------------------------|-----------|
| 5. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

● TRANSITION TO OEI-FLIGHT

- | | |
|-------------|-----------------------------|
| 3. Airspeed | – 60 KIAS (V _Y) |
|-------------|-----------------------------|

- | | |
|--------------------------------------|-----------|
| 4. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

5. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS

3.3.3 Single Engine Failure - Takeoff

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

- | | |
|---------------------|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
|---------------------|---------------------------------|

● REJECTED TAKEOFF

- | | |
|---------------------|-------------|
| 2. Landing attitude | – Establish |
|---------------------|-------------|

- | | |
|---------------------|--|
| 3. Collective lever | – Raise as necessary to stop descent and cushion landing |
|---------------------|--|

After landing:

- | | |
|--------------------------------------|-----------|
| 4. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

● TRANSITION TO OEI-FLIGHT

- | | |
|-------------|-----------------------------|
| 2. Airspeed | – 60 KIAS (V _Y) |
|-------------|-----------------------------|

- | | |
|--------------------------------------|-----------|
| 3. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

4. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS

3.3.4 Single Engine Failure - Flight

Conditions/Indications

- **RPM** warning (N₁ SPLIT) on

Affected engine:

- **GEN** warning light on
- Instruments indicate power loss

Procedure

- | | |
|--|--|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Airspeed | – 60 KIAS (V_Y) |
| 3. Single engine emergency shutdown | – Perform |
| 4. LAND AS SOON AS PRACTICABLE | |

ENGINE EMERGENCY CONDITIONS

3.3.5 Single Engine Landing

Conditions/Indications

One engine inoperative (OEI)

Procedure

LANDING APPROACH

1. Airspeed – 60 KIAS (V_Y)
2. Shallow approach – Establish

ON FINAL

3. Airspeed – 40 KIAS
4. Rate of descent – 300 fpm

TOUCHDOWN

5. Airspeed – Reduce, depending on power available
6. Landing attitude – Establish
7. Collective lever – Raise as necessary to stop descent and cushion landing

ENGINE EMERGENCY CONDITIONS

3.3.6 Single Engine Emergency Shutdown

NOTE Before performing an inflight single engine emergency shutdown, determine if the situation will allow for OEI flight.

Make certain that:

- the controls of the affected engine are selected, and
- the collective lever is adjusted to maintain the normal engine within OEI-limits.

Procedure

- | | |
|--|------------------|
| 1. Power lever (affected engine) | – IDLE, then OFF |
| 2. EMER FUEL VALVE switch (affected engine) | – CLOSE |
| 3. FUEL PUMPS/SUPPLY TANK switch (affected engine) | – Off |

ENGINE EMERGENCY CONDITIONS

3.3.7 Inflight Restart

NOTE An inflight restart may be attempted after a flameout or shutdown subject to the pilot's evaluation of the cause of flameout.

Procedure

CAUTION DO NOT ATTEMPT INFLIGHT RESTART IF CAUSE OF ENGINE FAILURE IS OBVIOUSLY MECHANICAL.

- | | |
|------------------------------|--|
| 1. Collective lever friction | – Adjust to maintain position of lever when released |
| 2. Electrical consumption | – Reduce |
| 3. Power lever | – OFF |
| 4. STARTER/GENERATOR sw | – Off |
| 5. EMERG FUEL VALVE sw | – OPEN |
| 6. FUEL PUMPS/SUPPLY TANK sw | – On |
| 7. IGNITION sw | – On |
| 8. STARTER/GENERATOR sw | – START and hold |

CAUTION ENGINE GAS PRODUCER SECTION MAY LOCKUP AFTER AN INFLIGHT FLAMEOUT OR SHUTDOWN. THEREFORE, EXCEPT DURING AN EMERGENCY, INFLIGHT RESTART SHOULD NOT BE ATTEMPTED DURING THE TIME PERIOD BETWEEN ONE MINUTE AND TEN MINUTES AFTER FLAMEOUT OR SHUTDOWN.

At 15% N_1 :

- | | |
|-------------------------|-------------------------------------|
| 9. Power lever | – IDLE |
| 10. TOT and N_1 | – Monitor closely, observe TOT peak |
| 11. N_2 increase | – Monitor |
| 12. Engine oil pressure | – Positive indication |

At 58% N_1 :

- | | |
|---------------------------------|---------------|
| 13. STARTER/GENERATOR sw | – Release |
| 14. IGNITION sw | – Off |
| 15. STARTER/GENERATOR sw | – GEN ON |
| 16. Power lever | – FLIGHT |
| 17. Electrical consumers | – As required |
| 18. LAND AS SOON AS PRACTICABLE | |

If restart is not successful:

- | | |
|--------------------------------------|-----------|
| 19. Single engine emergency shutdown | – Perform |
|--------------------------------------|-----------|

ENGINE EMERGENCY CONDITIONS

3.3.8 Engine Overspeed - Driveshaft Failure

Conditions/Indications

- **RPM** warning (N_1 SPLIT) on

Affected engine:

- Torque decreases to zero
- N_1 decreases
- N_2 increases above N_{R0}

Normal engine:

- Torque, N_1 and TOT increase
- N_{R0} and N_2 decrease

Procedure

- | | |
|--|--|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Affected engine | – Identify |
| 3. Single engine emergency shutdown | – Perform |
| 4. LAND AS SOON AS PRACTICABLE | |

ENGINE EMERGENCY CONDITIONS

3.3.9 Engine Overspeed - Governor Failure

Conditions/Indications

- **RPM** warning may come on
- N_{R0} and both N_2 increase

Affected engine:

- Torque, N_1 and TOT increase

Normal engine:

- Torque, N_1 and TOT decrease

Procedure

- | | |
|---|--|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Affected engine | – Identify |
| 3. Power lever (affected engine) | – Move towards IDLE until N_2 and N_{R0} stabilize in normal range |
| 4. LAND AS SOON AS PRACTICABLE | |

ENGINE EMERGENCY CONDITIONS

3.3.10 Engine Underspeed - Governor Failure

Conditions/Indications

- **RPM** warning may come on

Affected engine:

- Torque, N_1 and TOT decrease
- N_2 may decrease

Normal engine:

- Torque, N_1 and TOT increase
- N_{R0} and N_2 decrease

Procedure

- | | |
|---------------------------------------|--|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Affected engine | – Identify |
| 3. ENGINE TRIM switch | – Compensate N_2/N_{R0} drop |
| 4. LAND AS SOON AS PRACTICABLE | |

If no residual torque (affected engine) is available:

- | | |
|---|------------------|
| 5. Single engine emergency shut-down | – Perform |
|---|------------------|

ENGINE EMERGENCY CONDITIONS

3.3.11 Compressor Stall

Conditions/Indications

- Dull humming or popping sounds
- Torque and N_1 indications erratic
- TOT fluctuates
- Possible light yaw jerk

Procedure

1. Collective lever	– Adjust to OEI-limits or below
2. Affected engine	– Identify
3. Power lever	– Move slowly to IDLE

If compressor stall is still evident:

4. Single engine emergency shut-down	– Perform
5. LAND AS SOON AS PRACTICABLE	

If compressor stall ends:

- | | |
|----------------|-------------------------|
| 4. Power lever | – Move slowly to FLIGHT |
|----------------|-------------------------|

NOTE Severe stalls can cause damage to engine and drive system components and must be handled as an emergency condition.

Stalls of less severe nature (one or two low intensity pops) may permit continued operation of the engine at a reduced power level thus avoiding the condition that caused the compressor stall.

ENGINE EMERGENCY CONDITIONS

3.3.12 Droop Compensation Failure

Conditions/Indications

- N_{R0} and both N_2 decrease when collective lever is raised
- N_{R0} and both N_2 increase when collective lever is lowered

Procedure

NOTE Avoid large collective lever changes.

1. Collective lever
 - Maintain N_{R0} within limits
2. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS

3.3.13 Engine Oil Pressure Low

Conditions/Indications

Affected engine:

- Oil pressure indication 3.5 kp/cm² or less

Procedure

1. Single engine emergency shutdown – Perform
2. LAND AS SOON AS PRACTICABLE

ENGINE EMERGENCY CONDITIONS**3.3.14 Engine Oil Temperature High****Conditions/Indications**

Affected engine:

- Oil temperature indication 107 °C or more

Procedure

- | | |
|---|------------------|
| 1. Power (affected engine) | – Reduce |
| 2. Oil temperature (affected engine) | – Monitor |
| 3. LAND AS SOON AS PRACTICABLE | |

If oil temperature remains above limit:

- | | |
|---|------------------|
| 4. Single engine emergency shut-down | – Perform |
|---|------------------|

ENGINE EMERGENCY CONDITIONS

3.3.15 Double Engine Failure - Hover IGE

Conditions/Indications

- Left yawing motion
- N_{R0} and both N_2 decrease
- **RPM** warning (N_{R0} LOW) on
- Both **GEN** warning lights on
- Engine instruments (both engines) indicate power loss

Procedure

- | | |
|----------------------------|---|
| 1. Right pedal | – Apply as necessary to stop yaw |
| 2. Landing attitude | – Establish |
| 3. Collective lever | – Raise as necessary to cushion landing |

ENGINE EMERGENCY CONDITIONS

3.3.16 Double Engine Failure - Flight

Conditions/Indications

- Left yawing motion
- N_{R0} and both N_2 decrease
- **RPM** warning (N_{R0} LOW) on
- Both **GEN** warning lights on
- Engine instruments (both engines) indicate power loss

Procedure

- | | |
|-----------------|-----------|
| 1. Autorotation | – Perform |
|-----------------|-----------|

ENGINE EMERGENCY CONDITIONS**3.3.17 Double Engine Emergency Shutdown****Procedure****• ON GROUND**

- | | |
|--------------------------------------|---------|
| 1. Both power levers | – OFF |
| 2. Both EMERG FUEL VALVE sw | – CLOSE |
| 3. Both FUEL PUMPS/SUPPLY TANK
SW | – Off |
| 4. Battery sw | – Off |

• IN FLIGHT

- | | |
|--------------------------------------|---------|
| 1. Both power levers | – OFF |
| 2. Both EMERG FUEL VALVE sw | – CLOSE |
| 3. Both FUEL PUMPS/SUPPLY TANK
SW | – Off |

ENGINE EMERGENCY CONDITIONS

3.3.18 Autorotation

Procedure

1. Collective lever	– Full down, maintain N_{R0} within limits
----------------------------	--

2. Airspeed	– 75 KIAS recommended
--------------------	------------------------------

NOTE Max range airspeed 100 KIAS at 85% N_{R0}
 Min rate-of-descent airspeed 60 KIAS at 100% N_{R0}

3. Double engine emergency shut-down	– Perform
---	------------------

AT APPROX 100-FT AGL:

4. Flare attitude	– Establish (approx 15° to 25°) to reduce forward speed and rate of descent; control N_{R0}
-------------------	---

AT APPROX 6 TO 10 FT AGL:

5. Flare attitude	– Reduce to approx 5°
6. Heading	– Maintain
7. Collective lever	– Raise to stop descent and cushion landing

8. Battery switch	– Off
--------------------------	--------------

NOTE The appropriate values must be adjusted according to prevailing conditions of gross mass, wind and terrain.